Progress on the maximum size of an almost-equidistant set in $\mathbb{R}^5$: $f(5) \leq 17$, toward the exact value

A computer-assisted campaign, with certified non-realizability at 18, 19 and 20 points

August 5, 2026, third snapshot (level 17: 12615/12654; controls suite passed)

Background

A finite point set in $\mathbb{R}^d$ is *almost equidistant* if among any three of its points some two are at unit distance; $f(d)$ denotes the maximum size of such a set. The known values are $f(1) = 4$, $f(2) = 7$, $f(3) = 10$ [4, 1] and $f(4) = 12$ (computer-assisted resolution of Conjecture 1 of [1]; see the accompanying note `f4_equals_12.pdf` in this repository). For $d = 5$ the published state of the art is

$$16 \;\leq\; f(5) \;\leq\; 20,$$

the lower bound by Larman–Rogers [2] (the odd half-cube $\{v \in \{\pm 1\}^5 : v \text{ has an odd number of } {+1}\text{s}\}/\sqrt{8}$, whose non-unit-distance graph is the Clebsch graph), the upper bound by Balko, Pór, Scheucher, Swanepoel and Valtr [1]. We found no later improvement in the literature (2020–2026); the asymptotic bound is $f(d) = O(d^{4/3})$ [3].

This note records the state of an ongoing computer-assisted campaign to determine $f(5)$ exactly, run with the same certified machinery that proved $f(4) = 12$, generalized from $\mathbb{R}^4$ to $\mathbb{R}^5$. All code and data are in this repository.

Result. $f(5) \leq 17$. *More precisely: there is no almost-equidistant set of* 18 *points in* $\mathbb{R}^5$ *(and none of* 19 *or* 20 *points), by certified elimination of all candidate unit-distance graphs at each of the three levels. Combined with the Larman–Rogers set,* $16 \leq f(5) \leq 17$.

Work in progress. *One level remains: of the* 12654 *candidate graphs on* 17 *vertices,* 12615 *are already certified non-realizable at the time of this snapshot (*12518 *of them by circle-free searches taking six seconds in total); the remaining* 39*, all with a circle stage, exhibit narrow arcs of the circle parameter on which full-width interval evaluation blows up — direct probes show every such arc dies within* ∼65 *search nodes once split, so these are scheduling casualties, not geometric obstructions, and are queued for a re-run with direct fine tiling (`results_n17.json` is updated continuously). The full validation-control suite passes on the final kernel build (kill controls* K_7 *and* $K_1{+}K_{2,2,2,2,2}$ *die in* 1–2 *nodes; the sliced cross-polytope survive control isolates the true mirror realizations in exactly the* 6/24 *slices containing them). Independent numerical optimization over* all 12654 *candidates (Levenberg–Marquardt, up to* 300 *restarts per graph) found no graph anywhere near realizable: the smallest residual is* 0.409*, against* 10^{-30} *for realizable controls. If every* 17*-vertex candidate is eliminated,* $f(5) = 16$*; a certified survivor would instead pin* $f(5)$ *at* 17.

Reduction

Following [1], if $P \subset \mathbb{R}^5$ is almost equidistant then its unit-distance graph G satisfies: (i) $\alpha(G) \leq 2$ (the complement is triangle-free — this is almost-equidistance itself); (ii) G has no K_7 (at most $d{+}1$ points of $\mathbb{R}^d$ are pairwise at unit distance); (iii) G has no $K_{3,3,3}$ subgraph ([1], Lemma 11: for odd d, $K_{(d+1)/2}(3, \ldots, 3)$ is not a unit-distance graph in $\mathbb{R}^d$; three 3-point classes with all cross distances 1 would need three pairwise orthogonal $\geq$ 2-dimensional affine hulls, i.e. dimension ≥ 6). Graphs with (i)–(iii) are the *abstract almost-equidistant graphs in* $\mathbb{R}^5$.

Deleting edges of G that are removable without violating (i) yields a spanning *minimal* abstract almost-equidistant subgraph G' ((ii),(iii) survive edge deletion). Any realization of G restricts to one of G': n distinct points with every G'-edge at distance exactly 1 (non-edges unconstrained). Hence: *if no minimal abstract almost-equidistant graph on n vertices is realizable with n distinct points, then $f(5) < n$.* Minimal graphs are exactly the complements of *maximal triangle-free* graphs satisfying (ii) and (iii).

Candidate generation, independently verified

Maximal triangle-free graphs were generated with `triangleramsey` [5, 6] and filtered by our own checkers for (ii) and (iii) (`filter_mtf.c`); every input graph is re-verified to be maximal triangle-free from scratch. The resulting counts reproduce [1] (Table 2, $d = 5$ column) exactly:

n	17	18	19	20	21
maximal triangle-free graphs	164 796	1 337 848	13 734 745	178 587 364	2 911 304 940
minimal abstract a.e. graphs	12 654	8 825	340	8	0

The $n = 21$ row is an independent re-verification of $f(5) \leq 20$ [1]: all 2.9 billion maximal triangle-free graphs on 21 vertices were regenerated and none passes the filter.

As an independent check of the entire pipeline, a from-scratch enumerator (`enumaeq5.c`, growing graphs vertex-by-vertex with incremental tests of (i)–(iii), minimum-degree normalization, and isomorphism rejection — the method of Appendix A of the $f(4)$ note) reproduces the $d = 5$ column of [1] Table 3 at every order it reaches (7, 14, 38, 106, 402, 1817, 11 132, 86 053, 803 299, 7 623 096 for $n = 4, \ldots, 13$), and its minimal counts $(2, 3, 4, 6, 9, 14, 25, 46, 106, 242)$ match Table 2. At $n = 13$ the 242 minimal graphs from the two independent routes agree *as sets*, matched one-to-one up to isomorphism (`compare_sets.py`).

The lower bound is re-verified in integer arithmetic (`verify_lower_bound_16.py`): the 16 half-cube points are pairwise at squared distance 8 or 16 (unscaled), any three contain a pair at 8, and the non-unit graph is the 5-regular triangle-free Clebsch graph; scaling by $1/\sqrt{8}$ gives an almost-equidistant 16-point set, so $f(5) \geq 16$.

The certified decision engine in $\mathbb{R}^5$

`ckernel5.c` is a direct port of the certified $\mathbb{R}^4$ engine (`ckernel.c`; Appendix B of the $f(4)$ note), with the dimension-forced changes and nothing else:

- points are 5-dimensional interval vectors (outward-rounded IEEE doubles; cos / sin padded by 8 ulp);
- seeds are K_6 (canonical unit 5-simplex, exact interval enclosures, rigid) or K_5 (mirror explored by branching); 12 652 of the 12 654 graphs on 17 vertices contain K_6;
- a vertex with ≥ 5 placed neighbours is placed via the 5-sphere system (4 difference equations, 5 unknowns, 1-dimensional nullspace, quadratic — at most two roots), contracted by a 5×5 Krawczyk operator on the best-conditioned 5-subset of neighbours, all others acting as certified pruning constraints;
- a vertex with exactly 4 placed neighbours lies on a circle (3×3 interval Gram system for the circumcentre, 2-dimensional orthogonal complement basis by interval Gram–Schmidt), subdivided adaptively with the same runtime canary;

- near-tangent (merged-root) cases are resolved by the same one-dimensional segment subdivision;
- the two distinctness rules discard branches on which coincidence of non-adjacent vertices is forced: root separation now uses 5-subsets of common placed neighbours; the bisector-hyperplane rule needs 6 common placed neighbours with certifiably nonzero affine volume (5×5 interval determinant).

A branch is discarded only when an interval certifiably excludes a required value or when coincidence (fewer than n distinct points) is forced; eliminating all branches of one elimination order proves non-realizability with n distinct points. Elimination orders are generated circle-late; for the dense graphs at $n \geq 18$ almost all searches are circle-free binary trees (8813/8825 at $n = 18$; 339/340 at $n = 19$; 7/8 at $n = 20$), and every one of the 21 827 candidate graphs at levels 17–20 admits a valid order (no two-parameter stage is ever needed).

Controls. Verified at the time of this snapshot: the engine kills K_7 (no seven pairwise-unit points in $\mathbb{R}^5$) and does *not* kill the realizable controls K_6 (the unit 5-simplex) and $K_{2,2,2,2,2}$ (the unit cross-polytope, recovered numerically with distinct points and not killable by the engine within large node budgets). The remaining suite — the sliced survive controls for the cross-polytope and the 16-vertex half-cube graph, and the kill control $K_1 + K_{2,2,2,2,2}$ (an apex joined to the cross-polytope graph is non-realizable: the apex would have to be the centre, at distance $1/\sqrt{2} \neq 1$) — is running at snapshot time; its outcome will be recorded in `controls5_run.log` and in a revised note. (Survive controls run as sliced tilings and pass when slices report floor-width survivors; kill controls require every slice to die.)

Certified results at 20 and 19 points

For each graph, the driver (`reproduce5.py`) races the candidate elimination orders concurrently; a graph is certified when one order's tiling of the first circle parameter (a single unrestricted search for circle-free orders) is entirely killed.

- $n = 20$**:** all 8 minimal abstract almost-equidistant graphs certified non-realizable with 20 distinct points. Wall time ∼5 seconds total (Ryzen 9950X3D, LLVM clang build). Consequence: no 20-point almost-equidistant set in $\mathbb{R}^5$, i.e. $f(5) \leq 19$.
- $n = 19$**:** all 340 graphs certified non-realizable with 19 distinct points (∼170 seconds wall under load). Consequence: $f(5) \leq 18$.
- $n = 18$**:** all 8825 graphs certified non-realizable with 18 distinct points (8813 circle-free searches bulk-certified in ∼10 seconds of wall time on 20 workers; the 12 graphs with a circle stage certified by the racing driver in ∼1 second each). Consequence: $f(5) \leq 17$.

Per-graph verdicts and the winning decomposition indices are recorded in `results_n20.json`, `results_n19.json` and `results_n18.json`. Numerical corroboration: Levenberg–Marquardt finds best residuals ≥ 2.6 at $n = 19, 20$ and ≥ 0.4 over all of $n = 17$ (distinct-point solutions), i.e. nowhere near realizable.

Caveats

The proof of $f(5) \leq 18$ is computer-assisted, with the same trust base as the $f(4) = 12$ note: IEEE-754 correctly rounded arithmetic, library cos / sin accurate to 8 ulp (padded), the engine logic of Appendix B of that note with the $\mathbb{R}^5$ modifications above, and — for the *completeness* of the candidate lists — the `triangleramsey` generator [5], cross-validated here against [1] at $n = 17, \ldots, 21$ and against a fully independent enumeration up to $n = 13$ (counts at every order, sets at $n = 13$). The numerical sweeps are heuristic corroboration only; no conclusion rests on them. A reproduction run is one command per level (`python reproduce5.py -level 20`, etc.); the kernel compiles automatically and validation controls run first.

References

[1] M. Balko, A. Pór, M. Scheucher, K. Swanepoel, P. Valtr, *Almost-equidistant sets*, Graphs and Combinatorics **36** (2020), 729–754. arXiv:1706.06375.

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[4] B. Györey, *Diszkrét metrikus terek beágyazásai*, Master's thesis, Eötvös Loránd University, Budapest, 2004.

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[6] S. Brandt, G. Brinkmann, T. Harmuth, *The generation of maximal triangle-free graphs*, Graphs and Combinatorics **16** (2000), 149–157.